

Business Requirements Document

Hero Cycles Manufacturing Efficiency Dashboard

Project Overview

Hero Cycles is one of the world's largest bicycle manufacturers, operating large-scale production facilities with multiple machines, departments, supervisors, and production shifts. Efficient monitoring of production output, machine performance, downtime, and product quality is critical to achieving operational excellence.

The objective of this project is to develop a centralized Power BI dashboard that provides real-time visibility into manufacturing performance, quality management, and machine utilization. The dashboard should help stakeholders identify bottlenecks, improve efficiency, reduce downtime, and maintain product quality standards.

Business Objectives

The dashboard should enable stakeholders to:

- Monitor overall production performance.
 - Track machine efficiency and utilization.
 - Measure production downtime and identify causes.
 - Monitor quality performance and defect rates.
 - Evaluate supervisor and shift performance.
 - Identify underperforming machines and departments.
 - Improve operational efficiency and productivity.
 - Support data-driven manufacturing decisions.
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Stakeholders

Primary Users

- Plant Manager
- Production Managers
- Manufacturing Supervisors
- Operations Managers
- Quality Assurance Team

Secondary Users

- Maintenance Team
- Process Improvement Team

Dashboard Requirements

Page 1: Manufacturing Overview Dashboard

Purpose

Provide a high-level overview of manufacturing performance and operational efficiency.

Key KPIs

- Total Production
- Average Daily Output
- Total Downtime (Hours)
- Average Efficiency Rate (%)
- Machine Downtime (%)
- Defect Percentage (%)

Visual Requirements

Production Trend Analysis

Display monthly production trends to monitor manufacturing output.

Production by Machine

Compare production performance across machines.

Production by Shift

Analyze production contribution from different shifts.

Operational KPI Monitoring

Provide real-time visibility into production efficiency and downtime metrics.

Filters

- Year
 - Month
 - Machine Name
 - Supervisor
 - Product Type
 - Shift
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Page 2: Quality Management Dashboard

Purpose

Monitor product quality and identify areas contributing to manufacturing defects.

Key KPIs

- Total Production
- Total Defects
- Defect Percentage

Visual Requirements

Defect Analysis by Supervisor

Track defect counts and defect percentages by supervisor.

Defect Analysis by Machine

Identify machines contributing to higher defect rates.

Defect Analysis by Shift

Compare defect percentages across shifts.

Quality Performance Monitoring

Evaluate overall manufacturing quality trends.

Filters

- Year
 - Month
 - Machine Name
 - Supervisor
 - Product Type
 - Shift
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Page 3: Machine Utilization Dashboard

Purpose

Evaluate machine performance, efficiency, and downtime.

Key KPIs

- Average Efficiency Rate
- Total Downtime (Hours)

- Downtime Percentage

Visual Requirements

Efficiency by Machine

Compare machine efficiency across production lines.

Downtime by Machine

Identify machines with excessive downtime.

Downtime by Shift

Analyze downtime trends across production shifts.

Utilization Analysis

Measure overall machine performance and utilization.

Filters

- Year
 - Month
 - Machine Name
 - Supervisor
 - Product Type
 - Shift
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Page 4: Detailed Operational Data

Purpose

Provide transaction-level visibility into manufacturing activities for operational analysis and root-cause investigations.

Detailed Data Fields

- Date
- Day Name
- Machine Name
- Department
- Supervisor
- Category
- Category Description
- Product Name
- Shift Type
- Total Production
- Total Defects
- Defect Percentage

- Total Downtime (Hours)
- Downtime Percentage
- Average Efficiency Rate

Functional Requirements

- Export detailed records.
 - Drill-down analysis.
 - Operational troubleshooting.
 - Historical performance review.
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Business Questions Addressed

Production Analytics

- What is the total production output?
- How does production vary by month?
- Which machines contribute the highest production volume?
- Which shifts achieve the highest production output?

Quality Management

- What is the overall defect percentage?
- Which supervisors have the highest defect rates?
- Which machines generate the most defects?
- Which shifts experience the highest quality issues?

Machine Utilization

- Which machines have the highest efficiency rates?
- Which machines experience excessive downtime?
- How does downtime impact production performance?
- Which shifts demonstrate the highest utilization?

Operational Efficiency

- Which departments are operating most efficiently?
 - What factors contribute to reduced productivity?
 - Where should improvement initiatives be focused?
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Data Sources

Production Data

Contains:

- Production Date
- Machine Name
- Department
- Product Information
- Shift Information
- Production Quantity

Quality Data

Contains:

- Defect Quantity
- Defect Percentage
- Quality Categories
- Supervisor Information

Machine Performance Data

Contains:

- Machine Downtime
 - Downtime Percentage
 - Efficiency Rates
 - Utilization Metrics
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Data Model Requirements

The solution should follow a scalable analytical data model.

Fact Tables

- Production Fact Table
- Quality Fact Table
- Machine Performance Fact Table

Dimension Tables

- Date
- Machine
- Product
- Department

- Supervisor
- Shift

The model should support efficient filtering, drill-through analysis, and KPI calculations.

Success Criteria

The dashboard will be considered successful if it enables stakeholders to:

- Monitor manufacturing performance in real time.
 - Reduce production downtime.
 - Improve machine utilization.
 - Reduce product defects.
 - Increase production efficiency.
 - Improve quality monitoring.
 - Reduce manual reporting effort.
 - Support faster operational decision-making.
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Tools & Technologies

- Power BI
 - DAX
 - Power Query
 - Data Modeling
 - Microsoft Excel
 - Data Visualization
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Expected Business Impact

The Manufacturing Efficiency Dashboard will serve as a single source of truth for production, quality, and machine performance metrics. By providing real-time visibility into operational performance, the solution will help management identify bottlenecks, improve machine utilization, reduce downtime, minimize defects, and increase overall manufacturing efficiency. The dashboard will support continuous improvement initiatives and data-driven decision-making across manufacturing operations.